MRP Estimation of State-Level Trust in Public Schools

Overview:

My goal was to produce state-level estimates of institutional trust in public schools, using MRP to produce (hopefully) better estimates. I built the pipelines from 4 data sources:
- The YouGov baseline from 11/21/24 (with n of about 58k for this question)
- ACS 2023 (https://www.census.gov/programs-surveys/acs/microdata.html)
- CCES Cumulative from 2021-2024 (https://doi.org/10.7910/DVN/II2DB6)
- 2024 FEC Presidential Election Results (https://www.fec.gov/introduction-campaign-finance/election-results-and-voting-information/)

Due to the ACS survey not containing information on party ID, I used a synthetic joint distribution adapted from Shiro Kuriwaki's package (https://github.com/kuriwaki/synthjoint/). As outputs, I produced 3 sets of estimates, a baseline built without party ID from MRP on demographics alone, and then 2 variations with party ID using 2 different synthetic joint distribution methods.

Results:

I ran some comparisons between weighted survey mean and MRP, those results are at the bottom of this doc

Regression:

Here are the summary tables for the regressions w/ base and pid.

```
summary(base_model)
```

```
Loading required namespace: rstan

 Family: bernoulli
  Links: mu = logit
Formula: trust_schools_binary ~ sex_mrp + (1 | age_mrp) + (1 | educ_mrp) + (1 | race_mrp) + (1 |
STATE) + (1 | STATE:educ_mrp) + (1 | STATE:race_mrp)
   Data: rivers_cleaner_df (Number of observations: 62425)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~age_mrp (Number of levels: 4)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     0.32      0.20     0.12     0.84 1.00     2238     2960

~educ_mrp (Number of levels: 4)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     0.51      0.28     0.21     1.26 1.00     2083     2373

~race_mrp (Number of levels: 5)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     0.35      0.17     0.16     0.81 1.00     2045     2489

~STATE (Number of levels: 51)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     0.15      0.03     0.09     0.21 1.00     1391     1465

~STATE:educ_mrp (Number of levels: 204)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
```

```
sd(Intercept)     0.07      0.02     0.03     0.10 1.01      915     1071

~STATE:race_mrp (Number of levels: 251)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     0.11      0.02     0.07     0.15 1.00     1447     2227

Regression Coefficients:
             Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept       -0.48      0.34    -1.15     0.21 1.00     2296     2674
sex_mrpWoman     0.02      0.02    -0.01     0.05 1.00    10475     3255

Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
summary(pid_model)
```

```
 Family: bernoulli
  Links: mu = logit
Formula: trust_schools_binary ~ sex_mrp + pid3_mrp + (1 | age_mrp) + (1 | educ_mrp) + (1 |
race_mrp) + (1 | STATE) + (1 | STATE:educ_mrp) + (1 | STATE:race_mrp) + trump_z
   Data: rivers_cleaner_synth (Number of observations: 58484)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~age_mrp (Number of levels: 4)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     0.31      0.20     0.12     0.84 1.00     2022     2455

~educ_mrp (Number of levels: 4)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     0.38      0.25     0.14     1.08 1.00     1731     2538

~race_mrp (Number of levels: 5)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     0.31      0.17     0.14     0.74 1.00     1883     2440

~STATE (Number of levels: 51)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     0.13      0.03     0.06     0.20 1.01      764      589

~STATE:educ_mrp (Number of levels: 204)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     0.05      0.02     0.01     0.09 1.00      600      503

~STATE:race_mrp (Number of levels: 250)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     0.10      0.03     0.05     0.15 1.00      844     1445

Regression Coefficients:
                   Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept              0.27      0.31    -0.30     0.94 1.00     1852     2121
sex_mrpWoman          -0.14      0.02    -0.18    -0.11 1.00     7665     2883
pid3_mrpIndependent   -1.00      0.02    -1.05    -0.96 1.00     7038     3219
pid3_mrpRepublican    -1.54      0.02    -1.59    -1.49 1.00     7201     3300
trump_z                0.01      0.06    -0.10     0.13 1.00     3674     2696

Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
```

```
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Essentially, partisanship dominated. Republicans and Independents have massively negative log-odds compared to Democrats. The most major change between the baseline and pid was sex. Without pid, women were about 0, while after controlling for partisanship, they had -0.14 log odds.

One very interesting thing was that the vote share for Trump had essentially no effect, which suggests that red and blue states don't matter beyond individual party ID. Essentially, this would suggest a Republican in California is as roughly as likely to trust public schools as one in Alabama.

State estimates:

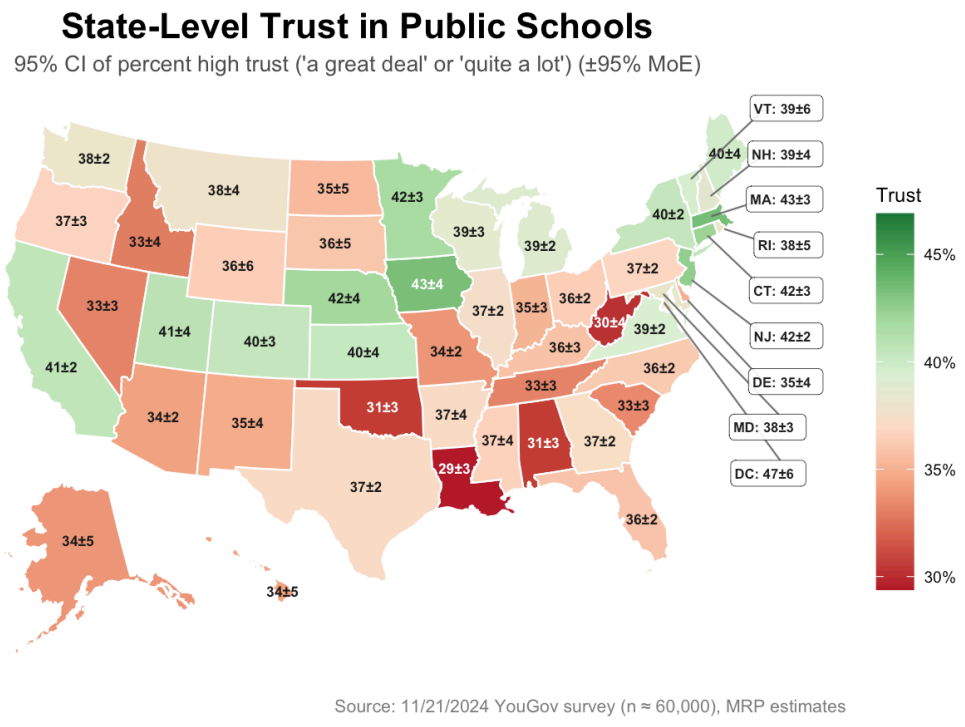
Table

Trust in Public Schools: MRP Estimates and Survey Means

State	BMLogit (PID)	Baseline	Survey (Raw)
D.C.	46.9% ± 6.0	42.1% ± 5.7	50.5%
Massachusetts	43.4% ± 2.7	43.4% ± 2.7	40.6%
Iowa	43.4% ± 3.5	44.2% ± 3.6	43.0%
New Jersey	42.5% ± 2.4	42.2% ± 2.6	42.7%
Connecticut	42.2% ± 3.2	42.9% ± 3.4	41.0%
Nebraska	41.9% ± 4.2	43.8% ± 4.3	41.5%
Minnesota	41.7% ± 2.6	43.1% ± 2.8	38.3%
Utah	40.8% ± 3.6	41.0% ± 3.8	41.0%
California	40.6% ± 1.6	40.2% ± 1.7	43.9%
Colorado	40.4% ± 2.6	40.6% ± 2.7	34.8%
New York	40.4% ± 1.6	40.9% ± 1.7	44.2%
Kansas	40.2% ± 3.6	42.2% ± 3.7	39.1%
Maine	39.8% ± 4.1	41.5% ± 4.4	33.5%
Vermont	39.5% ± 5.5	39.5% ± 5.7	37.7%
Virginia	39.2% ± 2.3	37.7% ± 2.3	35.9%
Michigan	39.0% ± 2.1	39.7% ± 2.2	35.1%
Wisconsin	38.6% ± 2.5	40.7% ± 2.6	33.0%
New Hampshire	38.6% ± 4.0	38.3% ± 4.3	31.4%
Rhode Island	38.5% ± 4.8	37.5% ± 5.0	40.2%
Maryland	38.2% ± 2.8	38.2% ± 2.9	34.7%
Washington	38.2% ± 2.3	39.7% ± 2.3	38.2%
Montana	37.9% ± 4.4	38.7% ± 4.8	26.0%
Illinois	37.5% ± 2.1	37.7% ± 2.1	37.1%
Georgia	37.3% ± 2.0	37.2% ± 2.2	39.8%
Texas	37.2% ± 1.7	36.9% ± 1.8	38.2%
Arkansas	37.0% ± 3.6	36.5% ± 3.5	35.6%
Pennsylvania	36.7% ± 1.8	37.0% ± 1.8	33.5%
Oregon	36.7% ± 2.6	37.7% ± 2.7	36.4%
Mississippi	36.6% ± 4.0	35.2% ± 4.3	32.8%
Wyoming	36.4% ± 5.7	36.9% ± 6.2	28.3%

State	BMLogit (PID)	Baseline	Survey (Raw)
Ohio	36.4% ± 1.9	37.2% ± 1.9	33.0%
North Carolina	36.1% ± 2.1	35.9% ± 2.2	30.8%
South Dakota	36.0% ± 5.1	39.0% ± 5.4	40.5%
Florida	35.9% ± 1.6	35.5% ± 1.6	35.0%
Kentucky	35.7% ± 2.9	37.2% ± 3.1	31.1%
North Dakota	35.4% ± 5.3	37.3% ± 5.9	38.8%
Indiana	35.1% ± 2.5	34.6% ± 2.7	33.9%
Delaware	35.1% ± 4.4	34.8% ± 4.6	36.0%
New Mexico	34.7% ± 3.8	33.2% ± 3.9	34.1%
Arizona	34.5% ± 2.3	34.0% ± 2.4	30.5%
Hawaii	34.3% ± 4.6	34.7% ± 5.2	31.3%
Alaska	33.9% ± 4.9	35.6% ± 5.2	46.2%
Missouri	33.9% ± 2.4	34.2% ± 2.5	30.7%
South Carolina	33.5% ± 2.7	33.2% ± 2.8	33.2%
Nevada	33.3% ± 3.3	32.2% ± 3.3	36.0%
Tennessee	33.2% ± 2.5	32.9% ± 2.6	28.6%
Idaho	33.0% ± 4.1	32.4% ± 4.2	26.8%
Alabama	30.5% ± 2.9	29.7% ± 3.0	29.8%
Oklahoma	30.5% ± 3.0	30.6% ± 3.1	23.2%
West Virginia	30.2% ± 3.7	31.9% ± 3.9	23.6%
Louisiana	29.4% ± 3.3	28.2% ± 3.4	23.1%

Map



Source: 11/21/2024 YouGov survey (n ≈ 60,000), MRP estimates

Data:

For institutional trust:

Trust in public schools binned into high (“A great deal” and “Quite a lot”) and low (“Some” and “Very little”), leaving 58,000 respondents.

For demographics:

- standard 4 group age (“Under 30”, “30-44”, “45-64”, “65+”)
- standard 4 group education (“HS or less”, “Some college”, “College grad”, “Postgrad”)
- 5 group race (“Hispanic”, “White”, “Black”, “Asian”, “Other”)
- sex/gender (“Man”, “Woman”)

For party ID:

- pid3 (“Democrat”, “Republican”, “Independent”) used from YouGov
- state marginals drawn from CCES pooled across 2021-2024

State context:

To differentiate pid3 and stablize across states, I wanted to add a secondary variable at the state level to account for differences in party identification at state level, so I added the z-score of the Trump 2024 vote share.

Methods:

Synthetic joint:

ACS has no partisanship, so had to use a synthetic joint distribution to poststratify on it. There are a few different ways of doing this, but they tend to either use an outside data source to find marginals or rely solely on the survey itself. To correct for the sampling bias of the YouGov survey, I used weighted CCES with its representative pid category as the marginals.

The two best synthetic joint methods for data of this scale are smoothfix, which rakes the ACS uniformly to hit CCES partisan marginals (which of course assumes independence) and bmlogit, a multinomial model which uses the survey data to learn partisan to demographic relationships and then reweights those to match CCES marginals. I decided to run both and the answer was they were essentially the same w/ r > 0.99, though bmlogit is methodologically cleaner. Here are the correlations at the cell level and in the final mrp results:

```
# Cell level
cor(frame_bm$count, frame_smfx$count)
```

```
[1] 0.991912
```

```
# Final MRP results
cor(mrp_bm$p_mrp_est, mrp_smfx$p_mrp_est)
```

```
[1] 0.9905192
```

Both were highly correlated with the (no pid) baseline, but not to the same degree.

```
# with pid vs. baseline
cor(mrp_bm$p_mrp_est, mrp_base$p_mrp_est)
```

```
[1] 0.9450402
```

MRP Model:

I adapted my workflow from Kuriwaki's ccesMRPrun (https://github.com/kuriwaki/ccesMRPrun) and ccesMRPprep (https://github.com/kuriwaki/ccesMRPprep) packages. For my regression formulas, I used:

```
# baseline
formula_baseline <- trust_schools_binary ~  sex_mrp +
  (1|age_mrp) + (1|educ_mrp) + (1|race_mrp) + (1|STATE) +
  (1 | STATE:educ_mrp) + (1 | STATE:race_mrp)

# with PID
formula_partisan <- trust_schools_binary ~  sex_mrp + pid3_mrp +
  (1|age_mrp) + (1|educ_mrp) + (1|race_mrp) + (1|STATE) +
  (1 | STATE:educ_mrp) + (1 | STATE:race_mrp) + trump_z
```

The size of the YouGov survey allows a couple of state-demographic interactions (so called "deep MRP") wihout overfitting (effective sample size remained fairly large). For the pid version, I used the z-score of Trump's 2024 vote share to capture potential state-level political party differences not fully accounted for by individual-level partisanship. The model converged fine w/ no divergent transitions and R-hat around 1.

Notes:

I did not include income due to already capturing education, but did run a linear regression on income for university and public school trust and found a somewhat modest negative effect smaller than education and significantly smaller than partisanship which ended up with a lot of bins and complicating the model.

I could also potentially run the MRP as ordinal over the 1-4 scale, which would be relatively easy to do if there are any notable individual states (though somewhat long and uncooperative over the whole country).

Validating:

Correlation between weighted survey data and MRP:

```
cor(mrp_bm$p_mrp_est, weighted_means$weighted_trust)
```

```
[1] 0.7376741
```

Shrinkage test, increases as expected with smaller state samples:

```
validation <- data.frame(
  STATE = weighted_means$STATE,
  n = weighted_means$n,
  diff = mrp_bm$p_mrp_est - weighted_means$weighted_trust)

cor(validation$n, abs(validation$diff))
```

```
[1] -0.2312922
```

Comparison plots:

Shrinkage diagnostic: MRP vs Weighted Survey

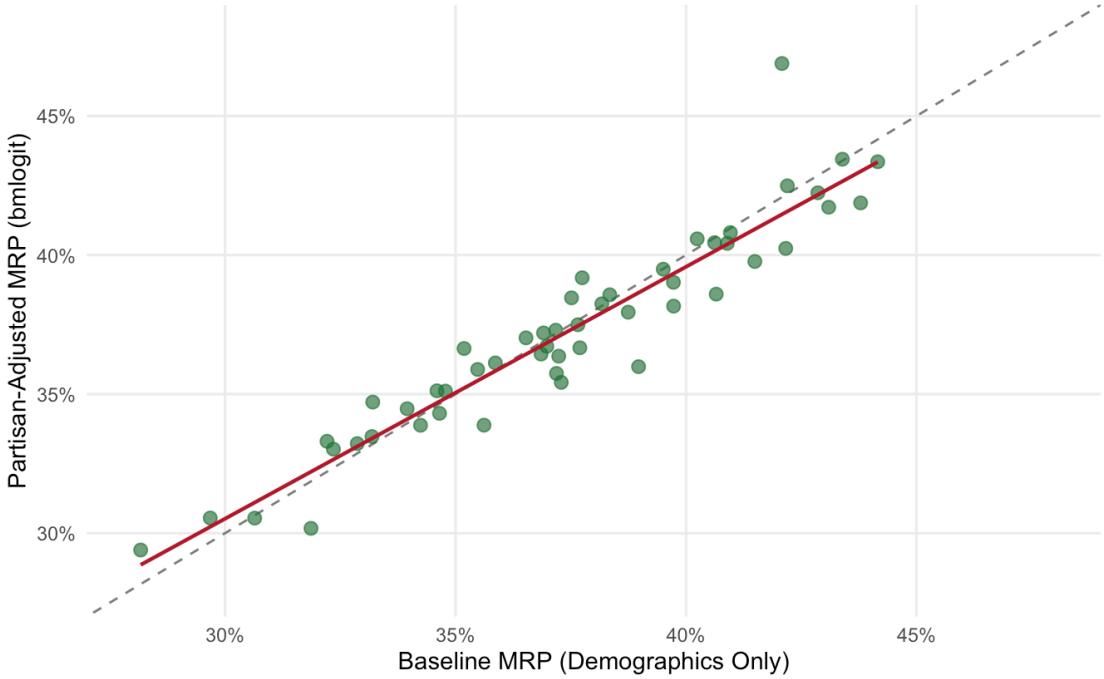
r = 0.74



Dashed line = perfect agreement; red line = linear fit

Effect of Partisan Adjustment on State Estimates

r = 0.95



Dashed line = no change from partisan adjustment